EE2015: Electric Circuits and Networks

Tutorial 6 Answer Key
(September 26, 2025)

1. (a) $i_L(0^+) = 0$, $v_R(0^+) = 4\,\text{V}$, $v_C(0^+) = -20\,\text{V}$
   (b) $i_L'(0^+) = 0$, $v_R'(0^+) = \frac{2}{3}\,\text{V/s}$, $i_R'(0^+) = \frac{1}{3}\,\text{A/s}$
   (c) $i_L(\infty) = 1\,\text{A}$, $v_R(\infty) = 4\,\text{V}$, $v_C(\infty) = -20\,\text{V}$

2. (a) $v_C(t) = 8\left(1 - e^{\frac{1}{4}t}\right)\text{V}$
   (b) $v_C(t) = 8\,e^{-\frac{t}{12}}\,\text{V}$

3. $i_x(t) = 0.8\,u(-t) + 0.8\,e^{-\frac{t}{480}}\,u(t)\,\text{A}$

4. $v_C(t) = \begin{cases} 8\left(1 - e^{-\frac{t}{5}}\right) & 0 < t < 1 \\ 1.45\,e^{-\frac{t-1}{5}} & t > 1 \end{cases}$

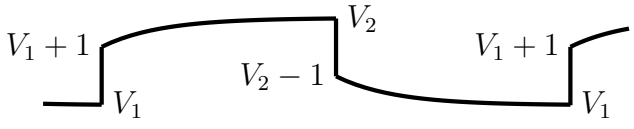
5. (a) $v_c(t) = e^{-t} - e^{-2t}$ for $t \geq 0$
   (b) $v_c(t) = t\,e^{-t}\,u(t)$

6. (a) $\dfrac{v_o}{v_i} = \dfrac{R_2}{R_1 + R_2} + \dfrac{R_1C_1 - R_2C_2}{(R_1 + R_2)(C_1 + C_2)}\,\exp\left(-\dfrac{t}{R_{\text{eq}}C_{\text{eq}}}\right)$,
   where $R_{\text{eq}}C_{\text{eq}} = (R_1 \parallel R_2) \cdot (C_1 \parallel C_2)$ and $v_i = V_p\,u(t)$
   (b) $v_o = V_p\left[1 - 2\,\exp\left(-\dfrac{t}{RC}\right)\right]$

7. (a)

   Switch at A: $i_L(t) = -3 + 5.09\,e^{-t/3}$
   Switch at B: $i_L(t) = -3 + 4.09\,e^{-t/2}$

   (b) $V_{\max} = 8.36\,\text{V}$, $V_{\min} = 4.36\,\text{V}$, $W_R = 216.3\,\text{J}$

8. 

   $V_2 = 0.623$
   $V_1 = -V_2$